Project | Sustainability Impact Analysis for Intel

INTRODUCTION: As you learned listening in on the strategy meeting with Dr. Alvarez and Intel's Sustainability Team, Intel is committed to reducing its carbon footprint and improving the sustainability of its devices – not just during manufacturing, but throughout the entire lifecycle.

A key part of this effort is their repurposing programs, which play a central role in achieving these sustainability goals. Repurposing and recycling programs aim to reduce e-waste, energy consumption, and CO_2 emissions by extending the life of existing devices, and thus reducing the need for new device manufacturing. Like Michael Campbell said: the average household in the US has anywhere from 3–5 PCs devices, tablets, notebooks, desktops that are perfectly functional, but not being used!

One challenge Intel faces is determining which devices in its repurposing program should be prioritized for the maximum environmental benefit. That's where data analysis comes in! To help with this, Intel gathered data on each device repurposed or recycled in 2024.

Your task is to evaluate the effectiveness of Intel's current repurposing strategy and provide a data-driven recommendation to help guide the program's direction and optimize sustainability efforts.

HOW IT WORKS: Follow the prompts in the questions below to investigate the data. Post your answers in the provided boxes: the **yellow boxes** for the queries you write and **blue boxes** for your text-based analysis. Once you're done, you'll submit your ***completed*** .pdf file to HQ for feedback from The Accelerator Team.

SQL App: Here's the link to our specialized SQL app, where you'll write your SQL queries and interact with the data.

NOTE: The dataset you are working with is designed for the Global Career Accelerator to reflect the key characteristics and structure of Intel's real data, while protecting their confidentiality and proprietary information. Be aware that any conclusions or results derived from this dataset should be viewed as hypothetical and for illustrative purposes only.

— Data Set **Descriptions**

In this project you'll query 2 different datasets, `intel.device_data` and `intel.impact_data`, that you will join together for your analysis. Here you'll find the data dictionary for each dataset.

`intel.device_data`

- `device_id`: Unique identifier for each repurposed device
- `device_type`: Type of device, values are either "Laptop" or "Desktop"
- `model_year`: The year the device was manufactured (e.g., 2018, 2019, etc.)

`intel.impact_data`

- `impact_id`: Unique identifier for the repurposed device's impact record (e.g., "LP20NA141592")
- `device_id`: Unique identifier linking the impact record to a specific device in the `intel.device_data` table
- `usage_purpose`: The specific purpose for which the device is being repurposed, values are Education & Digital Literacy, Corporate & Enterprise, Government & Public Sector, Environmental Sustainability Programs, and Social Impact & Non-Profit
- `power_consumption`: Power consumption of the device in watts (W) when in use (e.g., 50W, 75W)
- `energy_savings_yr`: Estimated energy savings per device per year when repurposed compared to a new device, measured in kilowatt-hours (kWh)
- `co2_saved_kg_yr`: Estimated CO2 emissions saved per device per year from manufacturing a new device, measured in kilograms (kg).
- `recycling_rate`: The percentage of the device that is recyclable (e.g., 80%, 90%).
- `region`: The geographical region where the device was repurposed, values are "North America", "Europe", and "Asia"

— **Task 1:** Organizing and Understanding the Data

We'll start by **joining** the device data with the impact data, allowing for a comprehensive analysis of device types, model years, repurpose regions, and energy savings in one dataset.

A. Simply write a query that returns all of the columns from both tables, joining the two on the `device_id` column. Be sure to choose the appropriate join so that all relevant

data is included in your result. **Note:** your query will have more than 150,000 rows (the max display for SQLPad!)

(paste your query below 👇)

```
SELECT
  d.*,
  i.*
FROM
  intel.device_data AS d
INNER JOIN intel.impact_data AS i
  ON d.device_id = i.device_id
```

B. To your joined dataset, add a new column called `device_age` calculated by subtracting the model_year from 2024. Paste your query below and double check that the values in your new column make sense. For example, a 2019 device should be 5 years old.

(paste your query below 👇)

```
SELECT
  d.*,
  i.*,
  (2024 - d.model_year ) AS device_age
FROM
  intel.device_data AS d
INNER JOIN intel.impact_data AS i
  ON d.device_id = i.device_id
```

C. Order your joined data by model_year (oldest to newest). Do you notice more older (5+ years) or newer (under 5 years) devices being repurposed? What might that indicate?

(write your **answer** below 👇)

Since we are only checking via ORDER BY, all the rows MY SQL was allowed to show was above the age of 5. But since we aren't using grouping to see the number of devices for each age, we can't be certain but from the given skim through of the database it suggests majority of the devices are old.

D. Bucketing the `device_age` will allow us to analyze trends and patterns in energy savings and CO_2 reductions more effectively than using individual ages. Use a `CASE WHEN` clause to add one more column, called `device_age_bucket`, to your data, that is based on the device_age:

- `WHEN` the `device_age` is less than or equal to 3, `device_age_bucket` should be "newer"
- `WHEN` the `device_age` is greater than 3 but less than or equal to 6, `device_age_bucket` should be "mid-age"
- `WHEN` the `device_age` is greater than 6, `device_age_bucket` should be "older"

HINT: Instead of using e.g. `device_age <= 3`, you need to reference the calculation directly: `2024 - d.model_year <= 3`.

Double check that the values in your new column make sense! For example, a 2019 device should be characterized as "mid-age".

(paste your query below 👇)

```
SELECT
  d.*,
  i.*,
  (2024 - d.model_year) AS device_age,
  CASE
    WHEN (2024 - d.model_year) <= 3 THEN 'newer'
    WHEN (2024 - d.model_year) >= 4 AND (2024 -
d.model_year) <= 6 THEN 'mid-age'
    WHEN (2024 - d.model_year) > 6 THEN 'older'
```

```
    ELSE 'ERROR'
  END AS device_age_bucket
FROM
  intel.device_data AS d
INNER JOIN intel.impact_data AS i
  ON d.device_id = i.device_id
ORDER BY device_age DESC
```

— **Task 2:** Key Insights

Now it's time to analyze the overall impact of Intel's repurposing program. You will use your final query from **Task 1** together with the `WITH` keyword for the remainder of this Project as you aggregate and analyze the data you've organized and prepped. For a refresher, rewatch "🍿The `WITH` Keyword" in SkillBuilder 6.

A. What is the total number of devices Intel repurposed in 2024?

HINT: The dataset **is** representing all devices repurposed in 2024! You just need to COUNT all the rows in your joined data from Task 1!

(write your **answer** below 👇)

601740 total repurposed devices in 2024

B. Write a query that returns the total number of devices repurposed, the average age of repurposed devices in 2024, the average estimated energy savings (kWh) from repurposed devices per year, and the total CO_2 emissions saved (in tons) from repurposed devices.

Note: CO_2 emissions are typically measured in tons. Since `CO₂_saved_kg_yr` is measured in kg, divide the `SUM(CO₂_saved_kg_yr)` by 1000 to report the total CO_2 emissions saved in tons.

(paste your query below 👇)

```
WITH joined_data AS (SELECT
  d.*,
  i.*,
  (2024 - d.model_year) AS device_age,
  CASE
    WHEN (2024 - d.model_year) <= 3 THEN 'newer'
    WHEN (2024 - d.model_year) >= 4 AND (2024 -
d.model_year) <= 6 THEN 'mid-age'
    WHEN (2024 - d.model_year) > 6 THEN 'older'
    ELSE 'ERROR'
  END AS device_age_bucket
FROM
  intel.device_data AS d
INNER JOIN intel.impact_data AS i
  ON d.device_id = i.device_id)

SELECT
  COUNT(*) AS num_repurposed_devices,
  ROUND(AVG(device_age), 2) AS avg_device_age,
  ROUND(AVG(energy_savings_yr), 2) AS
avg_energy_saving_yr,
  ROUND(SUM(co2_saved_kg_yr)/1000, 2) AS
total_co2_saved_tons_yr
FROM joined_data
```

C. Now that you have calculated the average estimated energy savings (kWh) and CO_2 emissions saved (tons), use ChatGPT to help put these numbers into perspective.

Try this prompt: I found that each repurposed device saves approximately of XXX kWh of energy per year and Intel's repurposing program saved XXX tons of CO_2 emissions in one year. Help me understand the significance of these numbers. How would this compare to the energy consumption of a small city or the amount of CO_2 produced by cars? What is the environmental impact of these savings?

What comparisons did you find most impactful in terms of scale? Summarize how much energy and CO_2 emissions were saved and how it compares to something familiar, like powering households or reducing car emissions.

(write your **answer** below 👇)

When comparing the energy consumption of a small city, every 35–40 repurposed devices can contribute to a single home's energy supply. Even though currently the number can't power a small city, with better efforts, that could be possible. Additionally, the CO2 reduction would be equivalent to taking 1,470 passenger cars off the road for a year. This shows how impactful the repurposing technology is for the environment.

— **Task 3:** Identifying Trends & Maximizing Sustainability

By grouping our data in different ways, we can uncover patterns in energy savings and CO_2 reductions. These insights will help us determine which categories of devices contribute the most to sustainability efforts and where Intel should focus its repurposing strategy for maximum impact.

A. Write a query that returns the total number of devices, the average energy savings, and the average CO_2 emissions saved (in tons), grouped by device_type.

Note (again): You'll need to divide `AVG(CO₂_saved_kg_yr)` by 1000 to report the average CO_2 emissions saved in tons.

(paste your query below 👇)

```
WITH joined_data AS (SELECT
  d.*,
  i.*,
  (2024 - d.model_year) AS device_age,
  CASE
    WHEN (2024 - d.model_year) <= 3 THEN 'newer'
    WHEN (2024 - d.model_year) >= 4 AND (2024 -
d.model_year) <= 6 THEN 'mid-age'
```

```
    WHEN (2024 - d.model_year) > 6 THEN 'older'
    ELSE 'ERROR'
  END AS device_age_bucket
FROM
  intel.device_data AS d
INNER JOIN intel.impact_data AS i
  ON d.device_id = i.device_id)
SELECT
  device_type,
  COUNT(*) AS num_repurposed_devices,
  ROUND(AVG(energy_savings_yr), 2) AS
avg_energy_saving_yr,
  ROUND(AVG(co2_saved_kg_yr)/1000, 2) AS
avg_co2_saved_tons_yr
FROM joined_data
GROUP BY device_type
```

B. Based on the results, which device type contributes the most to energy savings and CO_2 reduction? Why might that be the case?

Hint: Don't forget you can use ChatGPT as your Teammate to help think through your response!

(write your **answer** below 👇)

When looking at the average energy saving, laptops save slightly more than Desktops but the CO2 saving is pretty much identical. Furthermore, there are way more laptops being repurposed which would suggest that laptops have a larger impact on the environment for energy and CO2 saving.

C. Write a query that returns the total number of devices, the average energy savings, and the average CO_2 emissions saved (in tons), now grouped by `device_age_bucket`.

(paste your query below 👇)

```
WITH joined_data AS (SELECT
  d.*,
  i.*,
  (2024 - d.model_year) AS device_age,
  CASE
    WHEN (2024 - d.model_year) <= 3 THEN 'newer'
    WHEN (2024 - d.model_year) >= 4 AND (2024 -
d.model_year) <= 6 THEN 'mid-age'
    WHEN (2024 - d.model_year) > 6 THEN 'older'
    ELSE 'ERROR'
  END AS device_age_bucket
FROM
  intel.device_data AS d
INNER JOIN intel.impact_data AS i
  ON d.device_id = i.device_id)
SELECT
  device_age_bucket,
  COUNT(*) AS num_repurposed_devices,
  ROUND(AVG(energy_savings_yr), 2) AS
avg_energy_saving_yr,
  ROUND(AVG(co2_saved_kg_yr)/1000, 2) AS
avg_co2_saved_tons_yr
FROM joined_data
GROUP BY device_age_bucket
```

D. Based on the result of your query, what do you notice about the relationship between device age and the number of devices repurposed versus the average energy saved?

(write your **answer** below 👇)

There is an inverse relationship between the device age and the number of devices repurposed versus the average energy saved. In the query results, there are more newer and mid-age devices compared to the

older devices. being repurposed but on average they save less energy per year when compared to the older devices.

E. Finally, write a query that returns the total number of devices, the average energy savings, and the average CO_2 emissions saved (in tons), now grouped by region.

(paste your query below 👇)

```
WITH joined_data AS (SELECT
  d.*,
  i.*,
  (2024 - d.model_year) AS device_age,
  CASE
    WHEN (2024 - d.model_year) <= 3 THEN 'newer'
    WHEN (2024 - d.model_year) >= 4 AND (2024 -
d.model_year) <= 6 THEN 'mid-age'
    WHEN (2024 - d.model_year) > 6 THEN 'older'
    ELSE 'ERROR'
  END AS device_age_bucket
FROM
  intel.device_data AS d
INNER JOIN intel.impact_data AS i
  ON d.device_id = i.device_id)
SELECT
  region,
  COUNT(*) AS num_repurposed_devices,
  ROUND(AVG(energy_savings_yr), 2) AS
avg_energy_saving_yr,
  ROUND(AVG(co2_saved_kg_yr)/1000, 2) AS
avg_co2_saved_tons_yr
FROM joined_data
GROUP BY region
```

F. How does the carbon intensity of electricity in each region impact the total CO_2 savings from repurposed devices? Are there regions where repurposing leads to significantly higher environmental benefits? Why might that be?

(write your **answer** below 👇)

> Based on the query results, it is understandable that regions like Asia have a much higher average CO2 saving when compared to Europe and North America and this may be because there would be a higher carbon intensity of electricity in Asia and even though the energy saved is the same across all three regions, the environmental impact of CO2 saving is much higher.

— **Task 4:** Data-Driven Recommendations

Using the findings from this analysis, we need to summarize key takeaways and develop actionable recommendations for Intel. Remember: the goal is to refine Intel's repurposing strategy to maximize energy savings and CO_2 reductions while ensuring the most effective use of resources.

A. Based on your analysis of the repurposed devices (including energy savings, CO_2 emissions, and device age), write **four** key takeaways in succinct sentences/bullets that summarize the most important patterns and insights from the data. These should be specific, concise, and focused on the implications of repurposing newer versus older devices.

(write your **answer** below 👇)

> Laptops are the most repurposed device type (408064 devices), so they drive the largest total impact even if per-device savings are similar to desktops.
> Laptops also save the most energy per device (25.80), so they offer the highest per-device efficiency.
> Older devices (20239 devices) are repurposed less often, but they save the most energy on average (48.02), suggesting that the age increases the per-device environmental impact.Asia has the highest average CO2 savings per device (0.02 tons/year), likely due to the

higher grid carbon intensity, so repurposing there creates bigger environmental benefits per device.

B. Based on your four key takeaways and ChatGPT as your teammate, write a recommendation for Intel on how to improve the repurposing program. Your recommendation should include a clear action or strategy for Intel based on the data and a data-driven justification for why this approach would maximize energy savings and CO_2 reductions.

(write your **answer** below 👇)

Intel should improve the repurposing program for the Asia region and try to get as many older devices preferably repurposed. This is because older devices have a higher per-device energy savings and co2 saved and the Asia region has the higher per-device CO2 saved than the other regions. Additionally, considering that more than 50% of the population lives in Asia, there are way more devices we could receive back from the people to be repurposed.

C. Briefly reflect on how ChatGPT's suggestions influenced your recommendation. Did it help you see something you hadn't considered? What parts of your recommendation were improved based on its response?

(write your **answer** below 👇)

ChatGPT helps me choose a single main factor rather than a list of factors as a single well thought recommendation may be more appealing when listing recommendations.

— **LevelUp:** Optimizing Repurposing Strategy for Maximum Impact

Now that you've gained insights into the energy savings and CO_2 reductions across different device types and regions, let's use this data to optimize Intel's repurposing

strategy for maximum environmental benefit.

A. Add to your final query of Task 3 that returns the total number of devices, the average energy savings, and the average CO_2 emissions saved (in tons), grouped by region, **the percentage** of the total energy savings and CO_2 reductions contributed by each device type within each region.

HINT: To calculate the percentage of the total energy savings, use this formula: `Total energy savings for the device type / Total energy savings for the region) * 100` You'll use a similar one for the percentage of the total CO_2 reductions.



Try this prompt: What's the best way to calculate the percentage of CO_2 reductions contributed by each device type in each region?

(paste your query below 👇)

```
WITH
  joined_data AS (
    SELECT
      d.*,
      i.*,
      (2024 - d.model_year) AS device_age,
      CASE
        WHEN (2024 - d.model_year) <= 3 THEN 'newer'
        WHEN (2024 - d.model_year) >= 4 AND (2024 -
d.model_year) <= 6 THEN 'mid-age'
        WHEN (2024 - d.model_year) > 6 THEN 'older'
        ELSE 'ERROR'
      END AS device_age_bucket
    FROM
      intel.device_data AS d
    INNER JOIN intel.impact_data AS i
      ON d.device_id = i.device_id),
```

```
  region_device AS (
    SELECT
      region,
      device_type,
      COUNT(*) AS num_repurposed_devices,
      ROUND(AVG(energy_savings_yr), 2) AS
avg_energy_saving_yr,
      ROUND(AVG(co2_saved_kg_yr)/1000, 2) AS
avg_co2_saved_tons_yr,
      ROUND(SUM(energy_savings_yr), 2) AS
total_energy_saving_yr,
      ROUND(SUM(co2_saved_kg_yr)/1000, 2) AS
total_co2_saved_tons_yr
    FROM joined_data
    GROUP BY region, device_type),

  region_only AS (
    SELECT
      region,
      COUNT(*) AS num_repurposed_devices,
      ROUND(AVG(energy_savings_yr), 2) AS
avg_energy_saving_yr,
      ROUND(AVG(co2_saved_kg_yr)/1000, 2) AS
avg_co2_saved_tons_yr,
      ROUND(SUM(energy_savings_yr), 2) AS
total_energy_saving_yr,
      ROUND(SUM(co2_saved_kg_yr)/1000, 2) AS
total_co2_saved_tons_yr
    FROM joined_data
    GROUP BY region)

SELECT
  rd.region,
  rd.device_type,
  rd.num_repurposed_devices,
  rd.avg_energy_saving_yr,
  rd.avg_co2_saved_tons_yr,
```

```
  ROUND(rd.total_energy_saving_yr /
ro.total_energy_saving_yr * 100, 2) AS
pct_of_region_energy_saving,
  ROUND(rd.total_co2_saved_tons_yr /
ro.total_co2_saved_tons_yr * 100, 2) AS
pct_of_region_co2_saved
FROM region_device AS rd
INNER JOIN region_only AS ro
  ON rd.region = ro.region
ORDER BY
  rd.region,
  rd.device_type
```

B. Based on the results of your query, analyze the data to answer:

- Which device types in which regions contribute the most energy savings and CO_2 reductions relative to their numbers?
- How can this analysis help Intel prioritize specific device types in certain regions to maximize environmental benefits?

(write your **answer** below 👇)

In all of the regions, laptops contribute to the majority of the energy savings and the co2 saved is approximately 68% for every region. This suggests that Intel should continue to prioritise repurposing laptops as it would be more common in households to have one or more laptops. This will lead to a higher environmental benefit.

C. In addition to focusing on sustainability, imagine Intel needs to optimize for cost-effectiveness in their repurposing program. How might you adjust your query to incorporate cost data (e.g., cost per repurposed device)? What strategies could Intel use to balance sustainability goals with cost constraints?

(write your **answer** below 👇)

I would want to have a data where I can use the expense Intel covers for each repurposed device and then group them by region and device type to see which one is more sustainable while being cost-efficitive for Intel to move forward with. If certain models require more money to be repurposed, it would be better to scrap it for parts rather than sending it under the process for repurposing them.

— Evaluation Rubric

Unlike your Milestones that were evaluated largely based on your effort, the evaluation of your Portfolio Project will follow traditional evaluation methods, with tasks assessed for correctness and assigned point values accordingly.

Partial credit will be given where parts of this task are correct, even if other parts are incorrect or incomplete.

Task title	Max points
Task 1: Organizing and Understanding the Data	40
Task 2: Key Insights	25
Task 3: Identifying Trends & Maximizing Sustainability	60
Task 4: Data-Driven Recommendations	75
TOTAL POINTS:	200
LevelUp	
Optimizing Repurposing Strategy for Maximum Impact	20